ReviveCheck

Release attestation middleware for Portaldot V3.

Approvals that contracts enforce: a release owner commits a sealed report's hash to the ReleaseRegistry, reviewers approve that exact hash, and any V3 contract reads it through the ReviveCheckGuard library.

Evidence on-chain: the sealed report itself is stored through the native Remark module and retrieved from any node by its hashes.

A decision a maintainer can inspect: a policy evaluation against fresh finalized reads, with every check, reason and next action.

Portaldot Hacker House 2026, track Modular Cloud and Middleware. Deployed and recorded on the V3 testnet.

The maintainer's problem

A Portaldot wallet or DEX token-list maintainer publishes entries that wallets and DEX allow-lists read.

Where the approval lives today: a JSON token list, a spreadsheet or a chat thread. No contract can read it.

What was approved: "the release", with no record of what the reviewer saw.

The evidence: a screenshot, or a link whose content can change.

Checking the chain: someone opens an explorer and reads one interface.

Nothing tells them when it stopped holding.

The promise

Publish entries supported by the exact approved evidence.

Understand why an entry is no longer eligible.

Restore eligibility through a new review and explicit binding.

A policy states what a business action needs beyond the contract's answer: the right network and genesis, the subject, the plan fingerprint, supported runtimes, evidence age, an explicit trusted reviewer set, the consumer's listing and the stored report, each checked against fresh finalized reads.

The evaluation returns `valid`, `recheck-required`, `expired` or `unknown`. **Only valid enables**; an unavailable read is `unknown`, never a pass.

A new report needs a new approval and an explicit rebinding; history is kept.

Refusal

Release v1.0.0 copies the Ethereum default of 18 decimals; the chain reports 6 through both interfaces. Three places refuse it.

Where	Refusal	Record
Live check, both interfaces	`token.listing` failed: `decimals: listed 18, chain 6`. Remediation: publish a release whose entry uses the chain's decimals (6), then recheck.	Report `7f01ed85…`, testnet #163478
Catalog, under a valid approval	A configuration that claims 18 decimals is withheld: `entry-mismatch:decimals` (configured 18, approved 6). The policy evaluation itself is `valid`.	`public/catalog/testnet-v2-wrong-entry`, on the inputs captured at lifecycle step 10 and in CI
Contract, dry-run	`list` with 18 decimals reverts with `DecimalsMismatch` in the ReviveApi dry-run; nothing was signed.	Lifecycle step 4, owner, no transaction

Correction, review, listing

Steps 1 to 5 of the run recorded on 2026-10-25: release `pusd-catalog-v2` on VerifiedTokenList v2. Step 4 is the dry-run refusal on the previous slide.

Step	Action	Signer	Extrinsic	Block	Policy after the step	Entry
1	check pUSD v1.0.1 live, publish report `eb960cc7…`, store it on-chain	owner	`0x2489bd65…`	#187058	unbound	withheld
2	submit it as revision 1 of `pusd-catalog-v2`; the operator binds it	owner	`0x4d09c150…`	#187069	recheck-required: approval missing, not listed	withheld
3	approve revision 1	reviewer	`0xf90702ef…`	#187095	recheck-required: not listed	withheld
5	list pUSD on v2, 6 decimals	owner	`0xc1af7571…`	#187124	valid	**available**

Invalidation

Steps 6 to 8. A new submission opens a new revision and clears every decision, so nothing carries over silently. The approval alone does not move the binding.

Step	Action	Signer	Extrinsic	Block	Policy after the step	Entry
6	recheck live, publish report `1faef921…`, store it on-chain	owner	`0x80d46a86…`	#187145	valid	available
7	submit it as revision 2: the approval clears and v2 stops listing pUSD	owner	`0xd4818dec…`	#187165	recheck-required: revision superseded, approval missing, not listed	withheld
8	approve revision 2; the catalog still holds the revision-1 binding, so this changes nothing for the old binding	reviewer	`0x5af17433…`	#187181	recheck-required: revision superseded, not listed	withheld

Recovery and revocation

Steps 9 to 12. Every step with its receipt and its policy decision.

Step	Action	Signer	Extrinsic	Block	Policy after the step	Entry
9	bind the revision-2 report explicitly	operator	none (operator binding)	no transaction	recheck-required: not listed	withheld
10	list pUSD again on revision 2	owner	`0x435979c0…`	#187205	valid	available
11	request changes on revision 2: the contract un-lists pUSD	reviewer	`0xa1b8688b…`	#187230	recheck-required: approval revoked, not listed	withheld
12	approve revision 2 again	reviewer	`0x6cee3f89…`	#187271	valid	**available**

The decision screen

The Catalog decision page, `#/app/catalog`: the published entry against the approved report, the on-chain approval, the listing, the policy verdict with its reasons, evidence age against the policy clock, the checkpoints, the next action and the binding history.

Three facts are kept apart: the on-chain approval (the registry's decision), the recorded execution (finalized receipts, each linked to its block) and the current policy eligibility (an evaluation at a checkpoint and time).

Evaluate now runs the same evaluator the CLI and the Token Catalog use, in the browser, against the current finalized block. No wallet; nothing is signed.

Recorded evaluation `9dfc49fa…` at #187279: `valid`, 35 checks; the published entry equals the approved report `1faef921…` field by field.

Validity is point-in-time: evaluate again before each use.

Portaldot modules

Two views of one runtime: the same contract is reached as an EVM transaction or a native `revive.call`, and read back through both `eth_call` and `ReviveApi.call`.

Module	What it does here
Revive	Runs the contracts: ReleaseRegistry `0xe949b200edf070b9d425b206ed24dc8fe5b7d339` (#162962) and VerifiedTokenList v2 `0xedaf8a25bb583ad164a94191b762414277198df9` (#177354).
ReviveApi	Maps accounts (`account_id` and `address` must round-trip) and finalized blocks (`ethBlockHash`), so every read is checked through both interfaces at one finalized height.
Assets and its ERC-20 precompile	Two views of the token: asset 7777 (pUSD) at `0x00001E6100000000000000000000000010000`. Native Assets storage is compared with the precompile's supply, holder balance and decimals at one checkpoint; the address adapter is scoped to the observed spec 3002.
Remark	`remark.store` keeps the report bytes in the block; `Remark.Stored` carries the publisher and the blake2-256 of the bytes; retrieval recomputes the SHA-256. Reports stored at #164266, #164335, #187058 and #187145, about 0.75 tPOT each.

Reuse

One implementation, `lib/service.js`, behind the browser, the MCP server, REST, the JavaScript module and the CLI: thirteen read-only tools. Writes are dry-run, then signed in the user's own wallet.

The package declares nineteen `exports` specifiers (`.`, `./policy`, `./current`, `./decision`, `./catalog-release`, `./health` and more), a `files` whitelist and three `bin` commands: `revivecheck`, `revivecheck-mcp` and `revivecheck-catalog`, the release gate that writes a versioned catalog release only when the entry is permitted.

A packed artifact, `revivecheck-0.5.0.tgz` (SHA-256 `d8184943…`, 53 files), produced with `npm pack` from commit `e105dfe` and verified from an empty install, including the shipped release-gate starter; a second consumer installs it from the tarball with `npm ci`.

That consumer is a DEX allow-list in a separate repository with its own policy (hash `795942d6…`), six offline scenarios, 19 passing tests and a CI workflow that has not yet run; its live runs on 2026-10-25 allowed pUSD (`valid` at #191199) and prepared catalog release 1 at #192724, which a separate reader consumed. The repository is local and unpublished.

Author-run: **portability, not adoption**. No builder outside the project has installed or configured ReviveCheck; an external builder running it is the next gate.

Measured value

Equivalent checks both ways at identical captured checkpoints, reported as measured: no speed difference is established. The maintainer pilot, `docs/PILOT_PROTOCOL.md`, is **defined and not yet run**.

Measure	Baseline script over raw reads	ReviveCheck
Decisions: 5 scenarios on captured inputs, 3 live	8 of 8 as predefined; 0 false enables, 0 false refusals	8 of 8 as predefined; 0 false enables, 0 false refusals; the arms agreed in every scenario
Live repeat decision, two pairs	19.1 s and 17.1 s; 99 % of wall time waiting for the public nodes	17.3 s (faster) and 24.0 s (slower: more round trips, every value read through both interfaces)
Set-up, once; human time not measured	a 217-line script: 19 constants and 10 facts about the chain and the contracts	`npm ci` 4.9 s warm or 14.4 s empty cache; 25 policy fields, 17 catalog fields, one binding per revision
Second reader, from the artifact alone	5 of 6 required items, no next action; 0 of 4 extras	6 of 6 required items; 4 of 4 extras: integrity hash, trust basis, policy identity, role control

Status and boundaries

Established: records intact, deployed, listed, evaluated, lifecycle complete. Not established: independently adopted.

Two project-controlled signers: owner `5GH8LpSTf6sp31FeacbFE99E3b9NQb5E7xWHeBL8XqW3YhTt` and reviewer `5DtqFzALznihBAE5wWmWJ1owZcFbiUX6hYKcvAZyt3sSUDjZ`. Role separation on-chain; both accounts belong to the project.

Expiry, runtime change and node outage: simulated locally, not recorded on the testnet.

Portaldot V3 testnet, chain 420420777, runtime `portaldot-v3/3002`; last checkpoint #187279; 12 live steps with 10 finalized receipts.

Site https://revivecheck.axiqo.xyz/ · source https://github.com/a252937166/ReviveCheck · explorer https://scanner-testnet-v3.portaldot.dev/ · status record: public/release-status.json

Source commit **`d319f1c1a6fa8f587baaa0e49a815fea166d8343`**, rendered after `node scripts/release-status.js` reported `ok` on it.